

Customer Credit Risk Project

Project Planning and Final Summary

Step 1: Project Planning and Problem Framing

What is Data Analysis?

Data Analysis is the process of collecting, cleaning, transforming, and analyzing data to discover useful information and support decision-making.

Steps in a Data Science Project

- 1 Problem Understanding
- 2 Data Collection
- 3 Data Cleaning and Preprocessing
- 4 Exploratory Data Analysis
- 5 Feature Engineering
- 6 Model Building
- 7 Model Evaluation
- 8 Deployment and Monitoring

Machine Learning Problem

This project can be framed as a **Binary Classification** problem.

Target Variable: **default_flag**

- **default_flag = 0**: Customer did not default
- **default_flag = 1**: Customer defaulted on a loan

A future Machine Learning model can use customer demographics, financial information, loan details, transaction behavior, and engineered features to predict loan default.

Final Summary

Techniques Used

- Loaded data from CSV, JSON, and MySQL sources.
- Merged datasets using `customer_id` and `product_id`.
- Performed data understanding using Pandas.
- Generated a data profiling report.

- Handled missing values using Simple Imputer, Most Frequent Imputation, Random Sample Imputation, KNN Imputer, MICE, and Complete Case Analysis.
- Detected outliers using Z-score, IQR, and Percentile methods.
- Applied Winsorization to handle extreme values.
- Converted date variables and created date-based features.
- Applied Ordinal Encoding, Label Encoding, and One-Hot Encoding.
- Applied Binning, Binarization, Quantile Binning, and K-Means Binning.
- Applied StandardScaler, MinMaxScaler, MaxAbsScaler, RobustScaler, and Normalizer.
- Created new features such as Debt-to-Income Ratio and Average Monthly Transactions.
- Applied Log, Reciprocal, Square Root, Box-Cox, and Yeo-Johnson transformations.
- Used ColumnTransformer for applying different preprocessing steps to numerical and categorical variables.

Biggest Issues in Raw Data

- Missing values
- Outliers
- Multiple data sources
- Categorical variables
- Date and time variables
- Mixed data types
- Skewed numerical variables

Conclusion

The final dataset was collected from multiple sources, cleaned, transformed, encoded, scaled, and feature-engineered.

The processed dataset is ready for future Machine Learning Binary Classification to predict whether a customer is likely to default on a loan.